

NAMIBIA UNIVERSITY OF SCIENCE & TECHNOLOGY

Full Proposal for the Higher Degrees Committee

Proposed Title	The Role Of Natural Language Processing In Creating Ju/'hoansi Stories With Structurally Accurate Cultural Narratives
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Abstract

Natural Language Processing (NLP) has made significant strides in enabling machines to understand and generate human language. However, much of the existing research has focused on languages with abundant digital resources, leaving African indigenous languages, such as Ju/'hoansi, underrepresented. Ju/'hoansi is an endangered language with a rich oral storytelling tradition, but it lacks the digital infrastructure that would allow for effective application of current NLP techniques. This study explores how NLP can be used to create culturally accurate and meaningful stories for the Ju/'hoansi community. By fine-tuning a large language model using existing Ju/'hoansi narratives, the research aims to generate new stories that align with the community's traditional storytelling structures. The study will assess the quality and cultural authenticity of these AI-generated stories, using both automated and community-based evaluation methods. This work not only aims to advance the field of NLP in relation to African languages but also seeks to contribute to the preservation of Indigenous knowledge and cultural heritage, helping ensure that the Ju/'hoansi stories are maintained for future generations.

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Chapter 1

Introduction

Natural language processing (NLP) is a subfield of linguistics and computer science that uses artificial intelligence to enable machines to interpret and understand human language. Its emergence has simplified work in learning language semantics, syntax, and pragmatics through computational analysis on human language (Khurana et al., 2023; Sawicki et al., 2023). Addressing these factors has allowed NLP to expand to domains of translation and natural language generation, enabling it to bridge the gap caused by the learning curve needed to master or perfect a new language (Khurana et al., 2023).

NLP systems have particularly been useful in the African context; a substantial amount of research has gone into NLP for African languages, especially through translation models (machine translation) Lô et al. (2020). Large crawls of monolingual and parallel data of African texts have been scraped from the web, through which machine translation models have been trained to translate between hundreds of different languages Adelani et al. (2022). Though previous research has looked at NLP for language translation (Adelani et al., 2022; Lô et al., 2020), much work is still needed to use NLP to generate stories with structurally accurate African narratives. In particular, languages with little to no presence on the web that may be at risk of extinction (Biesele, 2006).

The challenge of limited data has a larger impact on indigenous communities; there is a greater level of under-representation in digital platforms and datasets for minority languages. Many African indigenous languages, such as Ju/'hoansi, have limited digital resources, such as annotated corpora, making mainstream NLP techniques effectively difficult to apply to these languages (cite). Additionally, African indigenous languages are deeply connected to unique cultural and narrative structures that differ from Western storytelling forms typically modelled by current language processing systems (Lô et al., 2020; Sawicki et al., 2023). These narrative styles, rich in

oral traditions, symbolism, and community context, are essential for any Generative AI system to generate meaningful indigenous stories (Lô et al., 2020). Ensuring that these cultural nuances are captured in language models is critical for producing narratives that truly reflect the experiences and world-views of indigenous communities.

1.1 Background to the study

Donkerbos is a settlement in the remote eastern part of Namibia in the Omaheke region, situated 260 km from the regional capital Gobabsis. The settlement is home to the Ju/hoansi, a San community that makes up most of the population of the settlement. Traditionally, the Ju/hoansi have occupied the hunting area around the settlement before the inception of the area. In the past Ju/hoansi were largely foragers exploiting over 100 species of plant food and 40 species of animals, indicating the rich indigenous knowledge they have amassed over the years (Wiessner, 2014). However since mid 1970s their culture has undergone a significant shift, now they have been settled in more permanent villages, such as Donkerbos, were they employ a mixed economy of foraging, income from waged labor and sales of crafts and government aid. (Wiessner, 2014).

Through a cultural shift that has been influenced by a myriad of factors, storytelling has remained deeply intertwined within the Ju/'hoansi culture. Campfire stories are a crucial part of Ju/'hoansi social life, they serve as a means to impart wisdom by the elders, maintain social harmony, and as a reflection that frames their understanding of life and the world around them (Keeney & Keeney, 2013; Wiessner, 2014). However, ju/'hoansi is only spoken by approximately 17000 people in Namibia and Botswana making it effectively an endangered language, rapid disappearance of the language is guaranteed if no action is taken (Jones & Biesele, 2018).

Over the years, collaborative efforts have been made through research and co-design to use emerging technologies to improve the social, educational, and economic well-being of the Donkerbos Ju/'hoansi san community (Kays et al., 2023). In particular, these efforts include resilience-building initiatives through existing and emerging technologies; the key is to involve community members to harness their personal stories and experiences (Kays et al., 2023). This project is part of a larger research initiative within the Faculty of Computing and Informatics at NUST, focussing on the preservation of indigenous knowledge. Under the auspices of the UNESCO Chair in Digital Technology Design with indigenous People at NUST, this research delves into the rich and contextualised stories and lessons from elders in the Ju/'hoansi community. By capturing and preserving these narratives, the project aims to extend the efforts to safeguard the indigenous Knowledge (IK) of the

Ju/'hoansi people.

1.2 Problem Statement

Language models are currently the most effective tool within the NLP toolbox, they do everything from translation to artistic story and poetry generation. However, they have limited access to African Indigenous knowledge and there isn't a lot of training done to make sure they are accurate with African languages. The lack of resources and annotated corpora of African text means that a lot of the popular language models, especially LLMs, have not been exposed to the nuances and intricacies of African text. Over the years, much effort has been made in other areas of NLP, such as Machine Translation (MT) and Neural Machine Translation (NMT) Adelani et al. (2022).

While Natural Language Processing (NLP) and large language models (LLMs) have made impressive strides, especially in creative areas like storytelling, they remain largely disconnected from Indigenous African languages such as Ju/'hoansi. Because these models have had little to no exposure to these languages, they struggle to capture the cultural frameworks, values, and storytelling traditions that are central to San communities Naveed et al. (2023).

This research tackles the challenge of using NLP to generate narratives that are both culturally meaningful and structurally faithful to a low-resource language like Ju/'hoansi — a language that existing LLMs have not been trained on.

Most current NLP work has centered around machine translation and text analysis for underrepresented languages. Very little has been done to create new, culturally rooted stories for languages that have limited digital footprints Adelani et al. (2022). Storytelling plays a crucial role in preserving the cultural identity of San communities, making it essential to develop approaches that do more than just translate words — approaches that can create new narratives in ways that honour Indigenous traditions.

In southern Africa, Ju/'hoansi is one such language, which is poorly documented but has a rich oral tradition. The project will focus on techniques for fine-tuning or steering language models with small, carefully curated collections of Ju/'hoansi stories. The goal is to produce narratives that are not only linguistically correct but also deeply respectful of the cultural heritage they represent.

1.3 Research Questions

This study seeks to answer the following research questions:

1. How can NLP techniques be adapted to generate stories with structurally accurate African narratives, particularly for the Ju/'hoansi community.?
2. What methods are suitable for fine-tuning NLP models to capture the linguistic nuances of the Ju/'hoansi language?
3. How can the generated stories be evaluated for relatability and cultural authenticity?

1.3.1 Main Objective

The main objective of this research is to fine-tune a large language model and develop an integrated web app powered by the underlying model to create new stories from existing stories within an Ju/hoansi San Community setting and context, particularly the San community in the Ju/'hoansi language.

1.3.2 Sub Objective

1. Develop metrics and qualitative criteria to evaluate the coherence, cultural authenticity, and narrative quality of ai stories generated.
2. Fine-tune a model to generate new stories from existing stories in the San language of Ju/'hoansi.
3. Benchmark AI generated stories against Ju/hoansi original stories on metrics of cultural authenticity and narrative quality.

1.4 Significance of research

In Southern Africa, Ju/'hoansi is a language that remains poorly documented despite its rich oral tradition. This research explores how Natural Language Processing (NLP) technologies can be leveraged to generate new stories from existing Ju/'hoansi narratives. Leveraging NLP in this way may help in preserving indigenous Knowledge and languages within Namibia. Structurally accurate narratives can serve as a relatable foundation for children and young adults to draw inspiration and motivation from when learning new skills (Foelske, 2014; Yoder-Wise & Kowalski, 2003). This is essential in a fast changing world, as people inherently anticipate learning from stories because that is where their earliest learning occured (Yoder-Wise & Kowalski, 2003). By creating new narratives, this study aims to help preserve the cultural heritage and linguistic diversity of the Ju/'hoansi people.
Additionally, this study contributes to the broader field of indigenous knowledge and language preservation. By developing indigenous knowledge systems, the study aids in ensuring that the wisdom and traditions of the Ju/'hoansi are not lost to future

generations. This research also underscores the potential of NLP in revitalizing endangered languages, such as Ju/'hoansi, and promoting cultural sustainability. Through these efforts, the study aims to empower the Ju/'hoansi community and support their cultural identity in the face of modern challenges.

Chapter 2

Literature Review

2.1 NLP and generative AI

Generative AI (Gen AI)is a subset of Artificial intelligence that focuses on creating content such as images, text and even music from data inputs and prompts Zhou and Lee (2024). In order to produce new and novel content, Generative AI employs algorithms trained on large datasets of existing content to predict the patterns of the content Guo and Chen (2024). Generative Adversarial Networks (GAN) and Variable Autoencoders are notable models in Gen AI, which have changed how content is created across various fields in digital art and content creation. Meanwhile, models like GPT and Llama have made significant strides in natural language generation and in producing coherent and contextually relevant data based on the scope of the large datasets on which they were trained.

This branch of AI is a subset of NLP, a branch in Artificial Intelligence and Linguistics that aims to make computers understand, interpret, and generate structured human language Khurana et al. (2023). NLP is classified into NLG (Natural Language Generation) and NLU (Natural Language Understanding)Khurana et al. (2023). These fields encompass a wide range of applications, including machine translation, which uses models to translate text from one language to another; sentiment analysis, which measures the emotional tone behind the text; and conversational agents, virtual assistants in the form of chatbots and personas like Alexa, Siri, Google Assistant.

Though there are many specialized applications of AI, as more applications become embedded with artificial intelligence, the interplay of NLP and Gen AI becomes more evident Sengar et al. (2024). For example, with each new iteration, tools such as GPT employ different models to cater to diverse use cases ranging from automated storytelling to code generation. Despite these advancements, both these fields face

challenges, including model bias, costly creation of quality datasets, privacy concerns, and ethical considerations of using realistic synthetic content and the potential misuse thereof Däubener et al. (2024). Tackling these issues to mitigate the social and technical implications is very important, as this will guide the frontier of how multi-modal AI systems are created, used, and improved.

2.2 The Role of NLP Techniques Preserving African Languages

A language can be described as a method-practice of broadcasting information, ideas, and emotions through a variation of a set of voluntarily produced symbols Khurana et al. (2023) and Sapir (2004). Languages play an essential role in safeguarding cultural heritage and social cohesion. Consequently, NLP tools can extend the reach of cultural heritage and knowledge (Traore & Diarra, 2023). For example, NLP-based translation models can be adapted to translate endangered languages and make them more accessible to speakers and learners. To this effect, applying NLP to protect and preserve African languages has garnered significant attention in recent years (Adelani et al., 2022; Lô et al., 2020).

NLP technologies such as translation, sentiment analysis, and Text-to-speech applications offer unique opportunities and challenges to the preservation of African languages(Adelani et al., 2022; Lô et al., 2020; Mohammed et al., 2023; Ndlovu & Moyo, 2023). Efforts have been made to save languages at the risk of cessation. For instance, Traore and Diarra, 2023 demonstrated how these technologies could save languages by aiding in language documentation and creating language learning tools. Additionally, as these technologies become more accessible, they enable greater involvement from marginalized communities. For example, Nokwary Technologies in Ghana has developed WhatsApp chatbots that local Ghanaians use for financial services (Ndlovu & Moyo, 2023).

On the other hand, African languages are diverse and numerous, which poses several challenges. Notably, the lack of prominence on the web for African languages at risk of cessation (Adelani et al., 2022; Traore & Diarra, 2023). While language models use large datasets that give them multilingual capabilities to generate new text in the context of existing material, they cannot effectively replicate these capabilities where some African text is concerned. In light of that, these languages have been classified as Low Resource Languages (LRL), and dataset creation of LRLs that are of substantial size and quality is often challenging to produce(Adelani et al., 2022; Lundby, 2008).

2.3 Status of the Ju/'hoansi language

Ju/'hoansi is notable for its complex phonetic system, which includes many consonants, vowels, and tonal variations. As a division in the Khoisan group of languages, it is one of the oldest languages in the world. Given that it is so old, this underscores the importance of the indigenous knowledge embedded into it. However, Ju/hoansi is classified as an endangered language, once primarily an oral language. However, due to social and economic change and the lack of consideration for indigenous languages in schools, the language is at risk of cessation (G/Ao, 2011).

The gradual decline of the language is partially due to the lack of usage by the younger generation; Ju/'hoansi students tend to struggle when they transition to mainstream Schools, even with early childhood education centres like the Nye Nye Village school (Davids, 2011). Another concern is that ju/hoansi. Traditionally, Ju/'hoansi has been an oral language carrying important lessons of cultural history and indigenous knowledge from one generation to the next. The Namibian government is trying to develop Ju/'hoansi for formal instruction to place it in schools along with other indigenous languages (Davids, 2011). However, these developments are limited, and there is still a need for a significant push to increase the formal support of Ju/'hoansi and other indigenous languages within the Namibian education system.

With the rise of AI and NLP systems, the future of the Ju/'hoansi language is not without hope. Continued research and innovation in the application of AI as a means of integrating storytelling and preservation has the potential to increase the yield of indigenous knowledge from the Ju/hoansi language. NLP integration could serve as a modern storyteller, enhancing traditional storytelling methods and ensuring that the rich oral traditions of the Ju/'hoansi are not lost (Inkingi, 2024). With an expanding catalogue of research in NLP for storytelling and new approaches to mitigate existing challenges within the field, there are many promising pathways to the preservation of the Ju/'hoansi language (Sudalairaj et al., 2024; Toussaint et al., 2024; Zhao et al., 2024).

2.4 African stories and AI

African narratives are shaped by diverse philosophical, political, historical, and linguistic traditions,(Eke & Ogoh, 2022). When viewed through the lens of African traditions, these forgotten narratives can guide how AI solutions are designed and developed in Africa. Eke and Ogoh (2022) makes the case that narratives ranging from Western non-contemporary to science fiction have influenced societies' perspectives on what AI can and should be. Therefore, guiding how it is designed and

implemented so far.

Concerning African storytelling, the storyteller engages with the audience through an interactive oral performance, ultimately ending with a moral lesson about everyday life Tuwe (2016). However, stories created by large language models do not follow the structure of traditional African stories. Instead, they are more closely aligned to Western narrative structures Lundby (2008). This phenomenon may be attributed to the limited training data in most NLP models due to language corpora, which are of limited quantity and quality Adelani et al. (2022). However, this further highlights the importance of having structurally accurate African narratives represented in the creation and use of AI systems.

There is a need to explore forgotten African narratives and harness these narratives. Creating and promoting content reflecting diverse African experiences and perspectives can enhance African Identity and visibility. Through the lens of Ghanaian folktales of Anansi and AI Inkingi (2024) explores these themes as well as cultural heritage and oral tradition and how these stories serve as a means to pass down cultural values (Ambala, 2020). This highlights the importance of integrating AI with African storytelling as a means to amplify African narratives by honoring rich cultural legacies and adapting to the day-to-day modern realities of Africans (Inkingi, 2024).

Chapter 3

Methodology

3.1 Study focus

This study builds upon previous research in NLP, which has predominantly focused on areas such as machine translation and neural machine translation. However, it acknowledges the limitations of existing research, particularly in the representation of African languages, which are often considered low-resource languages (Adelani et al., 2022). While previous efforts have addressed some aspects of African storytelling, the gap in generating stories that align with African narratives' structural and cultural nuances still needs to be filled.

3.2 Methodological Approach

The methodology used for this research will be research design. Design research involves finding innovative solutions to problems by combining prior knowledge, testing, and evaluating function and meaning (Faste & Faste, 2012). Research design leverages the study of technological phenomena to improve cultural systems all over the world (Faste & Faste, 2012). This makes it a suitable approach since this research project requires the exploration of a rapidly evolving technology and how it interacts with the nuances of the storytelling culture of the Ju/'hoansi community.

3.2.1 Research Phases

The research phases will follow the steps outlined by Wieringa (2014) in the design approach as follows:

Collection: Selecting stories from the digital repository to validate and understand the underlying narrative structure, as well as the role stories play within the Ju/'hoansi people, especially within the current social context. Narrative metrics will be devel-

oped to measure the general structure of the narrative. A diverse range of stories will be selected, representing different backgrounds and moral themes.

Identifying Ju/hoansi Stories and Narratives
The study will explore and seek out stories that best represent traditional Ju/'hoansi narratives. Categories of stories will be established with the help of the community through key individuals, such as elders or performers. As a starting point, the following categories will be collected:

- Origin Stories: Stories explaining how the world, animals, or humans came to be.
- Moral and Ethical Tales: Stories that teach lessons about behavior, survival, and community values.
- Animal Stories: Trickster tales involving characters like the hare or lion.
- Dream and Supernatural Stories: Stories about spirits, ancestors, and the unseen world.
- Conflict and Resolution Stories: Tales that showcase traditional dispute resolution or leadership.

Preserving Narrative Elements in Translation The study will attempt to prevent the loss of meaning and story structure by working with Ju/'hoansi speakers who understand both the language and storytelling traditions. The study will document oral performance elements and how they contribute to story structure. Back-translation techniques will be used (translate to English, then have another speaker translate it back to Ju/'hoansi to check accuracy).

Community Engagement The study will engage the Ju/'hoansi community, including key members who tell stories as well as those who listen and interact with them. Questionnaires with open-ended questions will be distributed, covering the following categories to guide the development of metrics for benchmarking against AI-generated stories:

- Describing the Ju/'hoansi storytelling tradition, identifying the types of stories that are told, their general structure, and how they are passed down through generations.
- Identifying the narrative structure of stories to pinpoint the key elements of a well-told story according to the Ju/'hoansi community. This section will also explore story patterns (beginning, middle, end), expressions, and rhythmic patterns in sections of stories.

- The role of stories in society, the lessons drawn from them, their role in everyday life, and whether certain stories are better suited for different ages or genders.
- General understanding and views on AI-generated stories. This will help gauge their perspective on AI-generated stories and how they would evaluate such stories if they were in the Ju/'hoansi tradition.

Design: Iteratively creating an effective solution to the identified problem while also developing an artifact that is both functional and relevant to the needs of its intended users. A model will be fine-tuned using the translated stories to effectively generate new, original Ju/'hoansi stories. A system will be built around the model to query the stories based on the context of a prompt (e.g., a background or scenario) so that the model can generate a story that matches the context and narrative structure of a typical Ju/'hoansi story.

System Design

The study will implement a technical solution architecture to guide the design and development of the system. Initially, Miro and Visio will be used to map out the system components, including data flow, model interactions, and user access points. The process will begin with training and fine-tuning a generative AI model using a cloud-based platform such as Azure ML or Google Vertex AI, where the dataset of Ju/'hoansi stories will be preprocessed and formatted for training. Next, the trained model will be integrated with a backend API, developed using FastAPI or .NET Web API, to allow for interaction with the model regardless of its hosting environment. To efficiently store and retrieve original Ju/'hoansi stories for comparison and benchmarking, a vector database such as Pinecone or Milvus will be used, enabling semantic similarity searches. The frontend will be built using .NET, providing a user-friendly interface where users can input prompts and retrieve AI-generated stories. Finally, the entire system will be deployed on a cloud platform such as **Azure, AWS, or Google Cloud, ensuring scalability, accessibility, and efficient model hosting.

Model Selection

A model will be selected from large language model (LLM) frameworks such as Hugging Face Transformers, OpenAI, and Meta AI. The study will utilize a pretrained language model that will be fine-tuned specifically for Ju/'hoansi storytelling. The selection process will consider factors such as model size, compute requirements, and adaptability to the dataset, ensuring an optimal balance between performance and feasibility.

Model Training and Development

The collected dataset will be formatted and tokenized for machine learning, ensuring that the model can effectively process and generate Ju/'hoansi stories. The selected pre-trained model will undergo fine-tuning using supervised learning, where it will learn to generate new stories based on provided prompts. The training process will involve adjusting key parameters such as batch size, epochs, and optimization algorithms to enhance performance while ensuring that the generated outputs maintain cultural relevance and narrative structure. Evaluation will be conducted through automated metrics—including fluency and coherence scores—to measure linguistic quality. Additionally, human assessments from Ju/'hoansi storytellers will be integrated to ensure that AI-generated stories align with traditional storytelling practices and cultural authenticity.

Knowledge Creation: This phase will focus on evaluating how well the developed system can generate Ju/'hoansi stories that reflect traditional storytelling patterns. The final artifact—a functional AI-powered application—will allow users to input prompts and receive AI-generated stories. These stories will be compared against original Ju/'hoansi stories stored in the digital repository, using the evaluation metrics developed during the model training and those established in the Donkerbos community.

The assessment will be both quantitative and qualitative. The study will use automated metrics to measure aspects such as fluency, coherence, structural integrity, and stylistic accuracy, ensuring that the generated stories retain key elements of Ju/'hoansi storytelling. Additionally, community evaluation will play a crucial role, with Ju/'hoansi community members reviewing the AI-generated stories based on the key metrics developed in 3 during the collection phase of the study. By comparing the similarities and differences between AI-generated and traditional stories, this study will answer the question of how well the model captures the essence of Ju/'hoansi storytelling.
Using the insights acquired, the study will detail the results of the findings and measure them against the key objectives and main research question established in the study.

3.3 Scope

The scope of this research project will be to create a story generation system which utilizes Natural language processing techniques to create new stories from existing narratives of the Ju/'hoansi language, which are structurally accurate. Proceeding the development of this system will be evaluating the stories against metrics to measure the authenticity and narrative quality of the stories generated. The scope will encompass the outlined objectives in the introductory chapter. 1

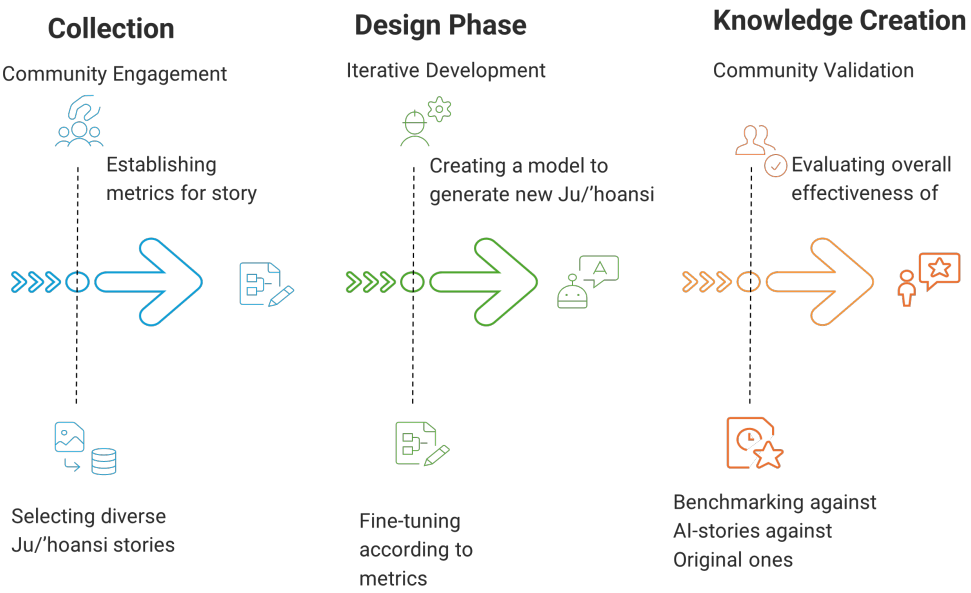


Figure 3.1: Project Phases

3.4 Delineation and Limitations

3.4.1 Limitations

Langauge barrier
This study will be conducted by individuals who do not speak the Ju/'hoansi language; therefore, it is crucial to find suitable translators who understand the nuances of the language. Additionally, during translation, context, expressions, and other communicative elements that rely on syntax and phonetic structure can often be lost.

Model maintenance
A model is only as good as its data; if not continuously fed with relevant data, its performance can degrade over time. A notable issue with Gen AI in text generation is that these systems can hallucinate and produce false information when not fine-tuned with recent text data. This means the developed system must be periodically retrained or fine-tuned with current data. Additionally, Reinforcement Learning from Human Feedback (RLHF) can be used to reduce hallucinations by incorporating human evaluations into the training process.

Community Engagement and Participation
The success of the project heavily relies on active participation and engagement from the Ju/'hoansi community. Any lack of involvement or support from the community could hinder the project's progress and outcomes.

Data Availability and Quality
Gathering sufficient and high-quality Ju/'hoansi stories in a format that can be used for training the NLP model may be a challenge. These stories will have to be well-documented and machine-readable in order to allow for effective model training.

3.5 Assumptions

A few key contextual and theoretical assumptions will guide this study. These assumptions help define the scope of the project and the expectations for its outcomes:

3.5.1 Representation of Indigenous Knowledge

The study assumes that the Ju/'hoansi narratives used will be rich and diverse enough to accurately represent the indigenous knowledge and cultural heritage of the community. The study also assumes that these stories will reflect the cultural values, resilience, strategies, and oral traditions of the Ju/'hoansi.

3.5.2 Robustness of Language Models

The research assumes that NLP models can be fine-tuned to capture the intricacies of African storytelling. The study also assumes that NLP models can be trained on a foreign language such as Ju/'hoansi such that they will be able to identify patterns within the data that can be used to reach the objectives of this study.

3.5.3 Willingness of Participants

The study assumes that once the necessary communication protocols have been handled, the participant will be active in voluntarily assisting the research effort through knowledge and story sharing.

3.6 Risk/Feasibility Analysis and Ethical Considerations

There has been a long-standing collaboration between the Namibia University of Science and Technology (NUST) and the Ju/'hoansi Donkerbos community. To ensure respectful and effective collaboration, the Donkerbos community will be engaged following the appropriate communication protocols that they set up.

The following ethical considerations will be taken into account for the duration of this research:

1. **Informed Consent and Participation**: Community members must be fully informed about the nature and potential outcomes of the project. Informed consent will be obtained to ensure that participation is voluntary. Continuous communication with community leaders and stakeholders will help ensure transparency and respect for their contributions.

2. **Preservation of Cultural Integrity**: The use of emerging technologies to generate new stories must respect and preserve the cultural integrity of the Ju/'hoansi people. The study will work closely with community elders to verify the authenticity and appropriateness of the generated narratives to avoid misrepresentation, appropriation.

3. **Ethical Use of Technology**: NLP and AI technologies can be misused, particularly in marginalised communities. To mitigate this, the study will implement safeguards to ensure the generated stories are used solely for educational and cultural activities, protecting the community from any potential exploitation.

4. **Data sovereignty**: Protecting the privacy and identity of the Ju/'hoansi community is critical. The Donkerbos community will own and give access to their data at their own discretion.

3.7 Work Plan

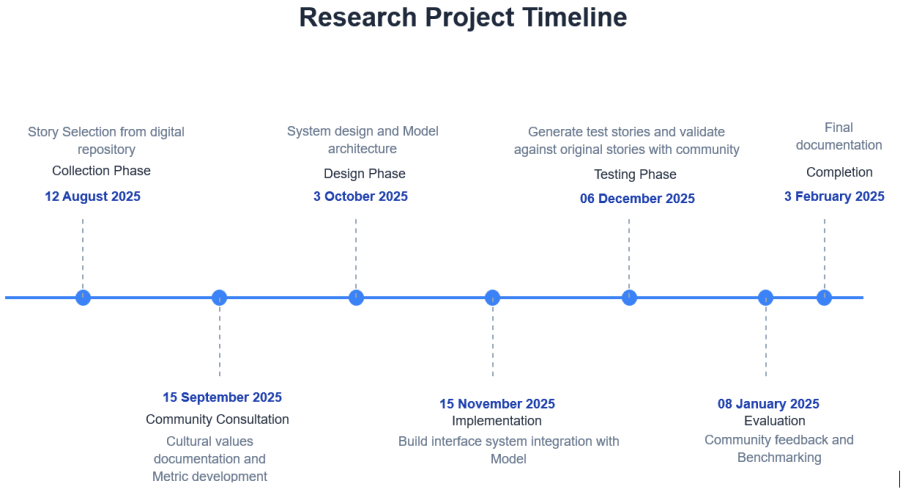


Figure 3.2: Artifact context interaction

3.8 Thesis structure

1. **Chapter One: Introduction:** This will introduce the topic and elaborate on the problem and proposed solution.
2. **Chapter Two: Literature review:** This will be a review of literature relating to and helpful for the research.
3. **Chapter Four: Research methodology:** This will outline the research methodology used.
4. **Chapter Five: Findings/results** The results or the research will be detailed here and explained in depth.
5. **Chapter Six: discussion** A further explanation and analysis of the results will be found in this section.
6. **Chapter Seven: Conclusions, recommendations and future works:** All ideas, methods and findings will be reiterated, summarised and discussed in this section of the paper.

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